
Comparative Study of WGAN-GP and Mini-Diffusion Models for Data Augmentation in Chest X-Ray Diagnosis

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1 Research Problem

Automated diagnosis of pneumonia from Chest X-Rays suffers significantly from class imbalance. In real-world clinical datasets, positive cases are often scarce compared to healthy controls [1]. This data bias causes deep learning classifiers to bias towards the majority class [2], leading to high false-negative rates, namely missed diagnoses, which is clinically unacceptable [3].

While data augmentation is a standard solution, traditional methods such as rotation, flipping, *etc.* fail to introduce new pathological variations. Generative AI offers a promising alternative; however, training state-of-the-art models, like Stable Diffusion, is computationally prohibitive for standard clinical workstations. Therefore, this research aims to solve the problem of **generating high-quality, pathologically accurate synthetic X-rays under computational constraints** to effectively augment training datasets.

2 Significance and Impact

Clinical Utility: Improving the Recall of diagnostic algorithms directly enhances patient safety by ensuring early detection. A generative augmentation tool can significantly boost model performance in hospitals with limited data.

Accessibility: By focusing on lightweight architectures, this research demonstrates that effective medical AI augmentation can be performed on consumer-grade hardware.

Privacy-Preserving Data Sharing: Synthetic data offers a viable pathway for sharing high-quality medical data without violating privacy regulations (e.g., Health Insurance Portability and Accountability Act, HIPAA).

3 Related Work

Data augmentation based on Generative Adversarial Networks (GANs) has been widely explored. [4] demonstrated that DCGAN-generated synthetic images improved CNN performance in liver lesion classification. Madani *et al.* [5] applied GANs to chest X-rays, proving generative models can capture anatomical structures. However, standard GANs often suffer from Mode Collapse and training instability. Gulrajani *et al.* [6] proposed WGAN-GP (Wasserstein GAN with Gradient Penalty) to mitigate these issues, yet GANs still struggle with fine-grained texture diversity.

Denosing Diffusion Probabilistic Models (DDPMs) have shown superior image quality and mode coverage compared to GANs [7]. However, their slow sampling speed and high memory consumption remain significant barriers for practical deployment. This project aims to bridge this gap by optimizing Diffusion models for medical image synthesis on smaller scales.

4 Methods

4.1 Dataset and Preprocessing

We use the Kaggle Chest X-Ray Images (Pneumonia) dataset [8]. Images are converted to grayscale, normalized to $[-1, 1]$, and resized to 64×64 for primary experiments. We will optionally attempt 128×128 if resources allow. Besides, we keep the official train and test split to avoid leakage. A resized chest X-ray image is shown in Figure 1.

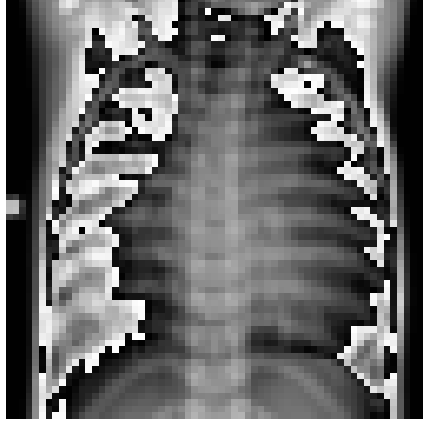


Figure 1: A training image compressed to 64×64

4.2 WGAN-GP

4.3 Model Architecture

The overall architecture of our model is illustrated in Figure 2. Following the guidelines of WGAN-GP [6], we design a generator G to map latent noise to synthetic chest X-rays, and a critic D to estimate the Wasserstein distance between real and generated distributions.

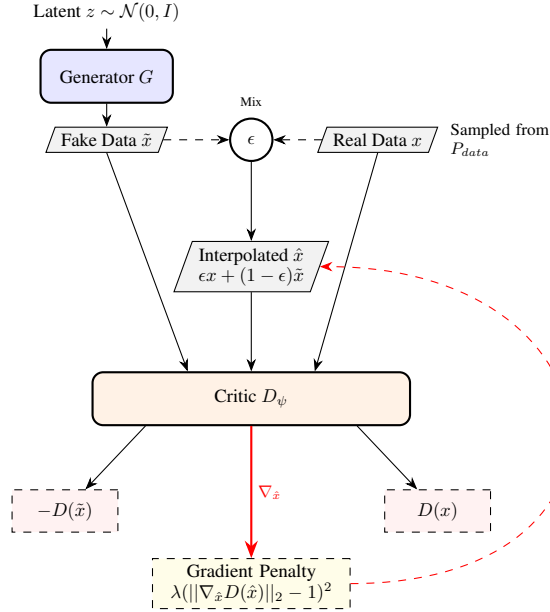


Figure 2: Overview of WGAN-GP architecture: The Generator G creates fake samples $\tilde{x}$. The Critic D evaluates real samples x , fake samples $\tilde{x}$, and interpolated samples $\hat{x}$. The gradient penalty (yellow box) is computed specifically on $\hat{x}$ to enforce the Lipschitz constraint.

4.3.1 Training Objective

Let $x \sim p_{\text{data}}$ be real images and $z \sim \mathcal{N}(0, I)$ be latent noise. The generator $G_\phi(z)$ produces synthetic images. The critic $D_\psi(x)$ estimates Wasserstein distance. WGAN-GP optimizes:

$$\max_{\psi} \mathbb{E}_{x \sim p_{\text{data}}} [D_\psi(x)] - \mathbb{E}_z [D_\psi(G_\phi(z))] - \lambda \mathbb{E}_{\hat{x}} (\|\nabla_{\hat{x}} D_\psi(\hat{x})\|_2 - 1)^2, \quad (1)$$

$$\min_{\phi} -\mathbb{E}_z [D_\psi(G_\phi(z))], \quad (2)$$

where $\hat{x} = \epsilon x + (1 - \epsilon)G_\phi(z)$ and $\epsilon \sim \mathcal{U}[0, 1]$.

4.3.2 Training Algorithm

We alternately update the critic and generator. The critic is trained for k steps to maximize the Wasserstein gap between real and generated samples, with a gradient penalty on interpolated samples to enforce the 1-Lipschitz constraint. The generator is then updated to increase the critic's score on generated images.

Algorithm 1 WGAN-GP Training

- 1: Initialize ϕ, ψ
 - 2: **for** each training iteration **do**
 - 3: **for** k critic steps **do**
 - 4: Sample real batch x , noise batch z , and $\epsilon \sim \mathcal{U}[0, 1]$
 - 5: $\tilde{x} \leftarrow G_\phi(z)$, $\hat{x} \leftarrow \epsilon x + (1 - \epsilon)\tilde{x}$
 - 6: $L_D \leftarrow -(\mathbb{E}[D_\psi(x)] - \mathbb{E}[D_\psi(\tilde{x})]) + \lambda \mathbb{E}(\|\nabla_{\hat{x}} D_\psi(\hat{x})\|_2 - 1)^2$
 - 7: Update $\psi \leftarrow \psi - \eta \nabla_\psi L_D$
 - 8: **end for**
 - 9: Sample noise z , $L_G \leftarrow -\mathbb{E}[D_\psi(G_\phi(z))]$
 - 10: Update $\phi \leftarrow \phi - \eta \nabla_\phi L_G$
 - 11: **end for**
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4.4 Mini-Diffusion Model

4.4.1 Model Architecture

Mini-Diffusion parameterizes the noise predictor $\epsilon_\theta(x_t, t)$ using a lightweight U-Net backbone. Given a noisy grayscale image $x_t \in \mathbb{R}^{1 \times H \times W}$ and timestep t , the network outputs $\hat{\epsilon} \in \mathbb{R}^{1 \times H \times W}$. We use sinusoidal timestep embedding followed by a small MLP, and inject the embedding into each residual block (additive and FiLM-style). To fit consumer GPUs, we adopt a reduced depth/width design and omit self-attention at 64×64 . The model architecture is shown in Figure 3.

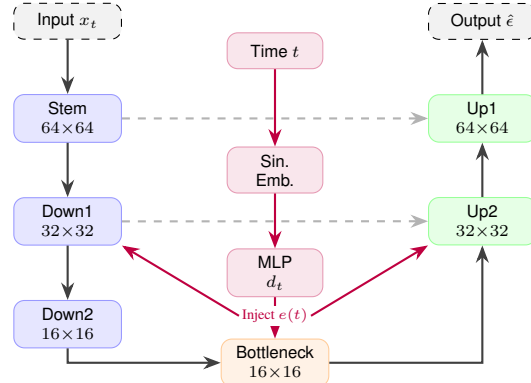


Figure 3: The Mini-Diffusion U-Net architecture. Pink modules are shifted upwards to interleave with skip connections.

4.4.2 Training Objective

We adopt the standard DDPM forward noising process:

$$q(x_t|x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I\right), \quad t = 1, \dots, T, \quad (3)$$

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad \bar{\alpha}_t = \prod_{i=1}^t (1 - \beta_i). \quad (4)$$

The model is trained to predict the injected noise via:

$$\mathcal{L}_{\text{diff}}(\theta) = \mathbb{E}_{x_0 \sim p_{\text{data}}, \epsilon \sim \mathcal{N}(0, I), t \sim \mathcal{U}\{1, \dots, T\}} \left[\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2 \right]. \quad (5)$$

4.4.3 Training and Sampling Algorithms

For generation, we run the learned reverse denoising process from $x_T \sim \mathcal{N}(0, I)$ down to x_0 . When using noise prediction, the DDPM mean can be written as:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right), \quad \alpha_t = 1 - \beta_t. \quad (6)$$

Algorithm 2 Mini-Diffusion Training (Noise Prediction)

- 1: Sample $x_0 \sim p_{\text{data}}$, timestep $t \sim \mathcal{U}\{1, \dots, T\}$, noise $\epsilon \sim \mathcal{N}(0, I)$
 - 2: Construct $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$
 - 3: Predict $\hat{\epsilon} = \epsilon_\theta(x_t, t)$
 - 4: Update θ by minimizing $\|\epsilon - \hat{\epsilon}\|_2^2$
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Algorithm 3 Mini-Diffusion Sampling (DDPM; optionally reduced steps S)

- 1: Initialize $x_T \sim \mathcal{N}(0, I)$
 - 2: **for** $t = T, T-1, \dots, 1$ **do**
 - 3: $\hat{\epsilon} \leftarrow \epsilon_\theta(x_t, t)$
 - 4: $\mu \leftarrow \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \hat{\epsilon} \right)$
 - 5: Sample $z \sim \mathcal{N}(0, I)$ if $t > 1$ else $z = 0$
 - 6: $x_{t-1} \leftarrow \mu + \sigma_t z$ (DDPM ancestral sampling)
 - 7: **end for**
 - 8: Return x_0
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To reduce inference cost, we optionally use a reduced number of denoising steps (e.g., $S \in \{25, 50\}$) via a strided timestep schedule / DDIM-style sampling), trading speed for quality.

5 Evaluation Protocol

5.1 Generative Quality: FID

We compute FID between real training images and generated images:

$$\text{FID} = \|\mu_r - \mu_g\|_2^2 + \text{Tr} \left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2} \right), \quad (7)$$

where (μ_r, Σ_r) and (μ_g, Σ_g) are feature statistics of real and generated sets (Inception features). However, FID uses Inception features pretrained on ImageNet (natural images), which may be imperfect for medical X-rays. We therefore report FID for comparability, but prioritize downstream classification and qualitative inspection for clinical relevance.

5.2 Downstream Utility: ResNet-18

We train a ResNet-18 classifier on:

- **Baseline:** real training set only;
- **Augmented:** real + synthetic images (controlled ratios such as +25%, +50%, +100% synthetic).

We evaluate on the official test split using AUC-ROC, F1-score, and Recall (priority for missed diagnosis reduction). We keep training hyperparameters constant across conditions to isolate the augmentation effect.

6 Discussion

In this project, we compare WGAN-GP and Mini-Diffusion under limited computational resources. We hypothesize that Mini-Diffusion can improve pneumonia class recall at the expense of longer inference time sampling, whereas WGAN-GP is fast but may produce repetitive samples.

6.1 Novelty

To compare WGAN-GP and Mini-Diffusion, we establish a controlled evaluation protocol that fixes the dataset, identical preprocessing and resolution, and the downstream ResNet-18 training configuration. In our project, attention modules are removed. Then we adopt a U-Net architecture and reduce the number of diffusion steps. Therefore, a lightweight diffusion model tailored for consumer GPU can be achieved. To evaluate this work, we assess not only the image fidelity with FID but also use AUC and recall to analyze the diagnostic utility.

6.2 Advantages

WGAN-GP offers fast inference speed and stable training objectives designed for limited computational training environments. However, due to mode collapse, WGAN-GP could generate repetitive images instead of capturing various pathological patterns.

In contrast, Mini-Diffusion displays more dynamics and mode coverage, which allows the generation of diverse and complicated structures. Nevertheless, the iterative denoising process requires longer inference time and higher computational cost compared to WGAN-GP.

As a result, the two models show a clear trade-off between generation speed and sample diversity, which will be further examined in subsequent experiments.

6.3 Limitations and Risks

Training at a 64×64 , or even 128×128 could fail to capture subtle textures, which impairs the downstream diagnostic performance. The FID originates from an inception network trained on natural images. However, chest X-rays belong to medical grayscale images. As a result, there is a huge difference between their distributions, which could not reflect diagnostic value of medical images.

References

- [1] Geert Litjens, Thijs Kooi, Babak Ehteshami Bejnordi, Arnaud Arindra Adiyoso Setio, Francesco Ciompi, Mohsen Ghafoorian, Jeroen AWM Van Der Laak, Bram Van Ginneken, and Clara I Sánchez. A survey on deep learning in medical image analysis. *Medical image analysis*, 42: 60–88, 2017.
- [2] Mateusz Buda, Atsuto Maki, and Maciej A Mazurowski. A systematic study of the class imbalance problem in convolutional neural networks. *Neural networks*, 106:249–259, 2018.
- [3] Luke Oakden-Rayner, Jared Dunnmon, Gustavo Carneiro, and Christopher Ré. Hidden stratification causes clinically meaningful failures in machine learning for medical imaging. *Proceedings of the ACM conference on health, inference, and learning*, pages 151–159, 2020.

- [4] Maayan Frid-Adar, Idit Diamant, Eyal Klang, Michal Amitai, Jacob Goldberger, and Hayit Greenspan. Gan-based synthetic medical image augmentation for increased cnn performance in liver lesion classification. *Neurocomputing*, 321:321–331, 2018.
- [5] Ali Madani, Mehdi Moradi, Alexandros Karargyris, and Tanveer Syeda-Mahmood. Chest x-ray generation and data augmentation for cardiovascular abnormality classification. In *Medical imaging 2018: Image processing*, volume 10574, pages 415–420. SPIE, 2018.
- [6] Ishaan Gulrajani, Faruk Ahmed, Martin Arjovsky, Vincent Dumoulin, and Aaron C Courville. Improved training of wasserstein gans. *Advances in neural information processing systems*, 30, 2017.
- [7] Walter HL Pinaya, Mark S Graham, Eric Kerfoot, Petru-Daniel Tudosiu, Jessica Dafflon, Virginia Fernandez, Pedro Sanchez, Julia Wolleb, Pedro F Da Costa, Ashay Patel, et al. Generative ai for medical imaging: extending the monai framework. *arXiv preprint arXiv:2307.15208*, 2023.
- [8] Daniel S Kermany, Michael Goldbaum, Wenjia Cai, Carolina CS Valentim, Huiying Liang, Sally L Baxter, Alex McKeown, Ge Yang, Xiaokang Wu, Fangbing Yan, et al. Identifying medical diagnoses and treatable diseases by image-based deep learning. *Cell*, 172(5):1122–1131, 2018. doi: 10.1016/j.cell.2018.02.010.